

WASP-11b

R-0717

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_241104 · created 2026-07-27T23:36:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460618.873 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.01
Transit depth (Rp/Rs)^2	1.5 +/- 0.26 [%]
Orbital Inclination (inc)	89.75 +/- 0.53
Airmass coefficient 1 (a1)	0.978 +/- 0.012
Airmass coefficient 2 (a2)	0.016 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#3 - [495, 331]
Optimal Method	PSF photometry
Transit Duration (day)	0.1058 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.122 ± 0.01 vs 0.131 (deviation 0.53σ)
Duration fit vs published	0.1058 d vs 0.1075 d (deviation 0.72σ)
Mid-time O–C	4.9 min
Depth SNR	7.2
Residual scatter	1.24 %

Light curve

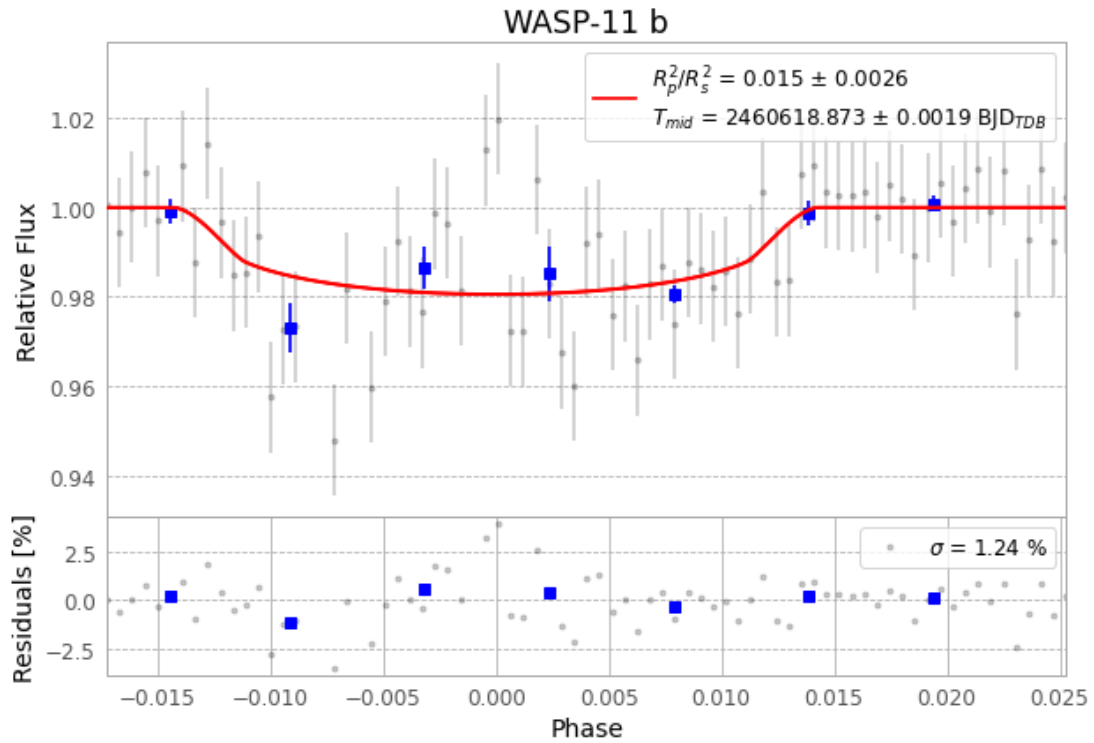


Figure 1 — FinalLightCurve_WASP-11 b_2024-11-04.png (SHA-256 65561e26451a2ad1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 287], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 61dabb24a78a11aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), HATP-10241104071516.FITS → HATP-10241104110317.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-241104.log (d86d98bf8698...), FinalLightCurve_WASP-11 b_2024-11-04.png (65561e26451a...), FinalLightCurve_WASP-11 b_2024-11-04.csv (c2464129ff97...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:36Z.